

Meromorphic periods, finite-contour L -functions, and the positive-weight Rademacher problem: context and novelty assessment

Working note (collaborator-facing)

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Abstract

This note records the full context around the de Rham–Betti (DRB) finite-contour construction of periods, quasi-periods, and L -functions for meromorphic modular forms, the original motivating goal (a positive-weight Rademacher-type coefficient formula), the precise analytic obstruction, and an end-to-end novelty assessment against the locally-harmonic-Maass-form (LHMF) / cycle-integral literature. The headline conclusions: (i) the “drag-the-contour-and-sum-residues” route reconstructs a known, divergent object for weight ≥ 0 ; (ii) the Duncan–Frenkel and DRB- s regularizations are both the wrong regulator for that divergence; (iii) the wall-crossing/chamber phenomenon is occupied ground (Bringmann–Kane–Kohnen; Löbrich–Schwagenscheidt; Choie–Zagier), but the finite-contour extraction of the completed all- s L -function together with de Rham–Betti quasi-periods appears transverse to that literature and is the defensible novel core.

1 The research program and its goal

The overarching aim is to relax the standard restriction “poles only at the cusps” to genuine interior poles in the upper half-plane $\mathbb{H}$, and to develop the analytic number theory (coefficients, L -functions, periods) of meromorphic modular forms with such poles.

The classical template is **Hardy–Ramanujan 1919** (the multiple-interior-poles result, distinct from their more frequently cited circle-method work on weakly holomorphic forms such as the partition generating function). HR 1919 produces exact, rapidly convergent expansions for the Fourier coefficients of forms like $1/E_6$, which is meromorphic of weight -6 with a pole at $\tau = i$ (where $E_6(i) = 0$) and at all Γ -translates of i ; the analogous $1/E_4$ has a pole at $\rho = e^{\pi i/3}$. These are properly meromorphic, interior-pole objects.

The program comprises:

- **The 1806 paper** (arXiv:1806.09874): L -functions for meromorphic modular forms regular at the cusps, built from a pole-subtraction (polylogarithm) step plus a deformed-Mellin contour. It explicitly flags the positive-weight Rademacher-coefficient problem as open.
- **The DRB note** (de Rham–Betti / Brown framework): a finite-contour cocycle construction yielding the periods $\omega^\pm(f)$, the quasi-periods $\eta^\pm(f)$, and an all- s completed L -function $L^*(f, s)$

for Δ , for the weakly holomorphic $\widehat{\Delta}$, and for general meromorphic f , with explicit wall-crossing across interior poles.

2 The objects and the drafts

The BFK L -function (the object DRB reproduces). “BKF” is Bringmann, Fricke, and Kent, *Special L -values and periods of weakly holomorphic modular forms*, Proc. Amer. Math. Soc. **142** (2014), 3425–3439 (the user dates the preprint ≈ 2011). It associates a regularized, incomplete-gamma completed L -series to weakly holomorphic modular forms and reads critical L -values off the error of modularity of an Eichler integral. It is widely cited (~ 50), but—per the user’s own audit of its citation tree—no citing work performs a *finite* (non-cusp, regularization-free) extraction of that L -function.

What DRB does to it. DRB integrates f against the polynomial kernel along a *finite* contour pair between interior basepoints and reads the cocycle off the result. The S -segment runs $-1/\tau_0 \rightarrow \tau_0$, the T -segment runs $\tau_0 - 1 \rightarrow \tau_0$; at the elliptic basepoint $\tau_0 = i$ the S -segment collapses and a single compact integral on $[i - 1, i]$ carries everything. The construction (a) reproduces the BFK completed $L^*(f, s)$ term by term in the Fourier expansion at every $s \in \mathbb{C}$, with numerical agreement to $\sim 10^{-53}$ for Δ and $\sim 10^{-47}$ for $\widehat{\Delta}$; (b) yields Brown’s periods $\omega^\pm(\Delta)$ and quasi-periods $\eta^\pm(\Delta)$ from a finite endpoint contour with no cusp divergence; and (c) makes “finiteness structural”: every integral is between interior points avoiding singularities, so there is nothing to regulate. The claimed value-add over BFK is therefore a *proof of correctness plus an explicit, numerically convergent reconstruction*, where BFK had a working-but-unexplained formula.

3 The original target and the analytic obstruction

Target. A convergent, Rademacher-type exact formula for the Fourier coefficients $c_f(n)$ of a *positive-weight* meromorphic modular form with interior poles, regular at the cusp—the natural generalization of HR 1919 past negative weight.

Status of the negative-weight case (solved). For weight < 0 the orbit sum converges and the problem is settled: Petersson’s meromorphic Poincaré series; Bringmann–Kane’s polar harmonic Maass forms give Hardy–Ramanujan-type expansions for all negative-weight meromorphic cusp forms bounded toward $i\infty$; HR 1919 itself handles $1/E_6$.

The crux obstruction (residue scaling). The literal “drag the contour to the cusp and sum the residues over the whole orbit” move reconstructs a Poincaré/Rademacher series whose convergence is governed by the modular transformation of residues. For f of weight k with a simple pole at τ_p , the residue at an image transforms as

$$\text{Res}_{\gamma\tau_p}(f) = (c\tau_p + d)^{k-2} \text{Res}_{\tau_p}(f), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Since $\#\{(c, d) : |c\tau_p + d| \leq X\} \sim X^2$, the orbit sum $\sum_{(c,d)} |c\tau_p + d|^{k-2}$ converges *iff* $k < 0$ and diverges for every $k \geq 0$. A weight-4 example is thus on the wrong side of the line; HR’s $1/E_6$ (weight -6) is on the easy side. This is exactly the obstruction named in 1806 (“when $k \geq 1$ they do not converge at all”).

Two convergence problems, not one. The polylogarithm subtraction (subtract $\sum_{\tau_p} \frac{1}{w_Q} \text{Li}_{1-p}(e(\tau - \alpha_Q))$ at each pole image) cures the *lower-strip* q -series convergence; it does nothing about the *orbit-sum* growth $|c\tau_p + d|^{k-2}$. Conflating the two is the source of the temptation. The weight-0 borderline is exactly where Bringmann–Kane’s “problem of Petersson” lives: the naive Poincaré series diverges and must be continued by Hecke’s trick, “further complicated by the fact that the Fourier expansion does not converge everywhere due to singularities in the upper half-plane.”

4 Two escape routes and their verdicts

(a) The DRB s -variable as a Hecke-trick regulator — does not work as posed. Hecke’s trick tames the orbit sum by inserting a damping factor $|c\tau + d|^{-2\nu}$ and continuing $\nu \rightarrow 0$. The DRB kernel supplies τ_p^{s-1} (and the Hurwitz-zeta T -kernel), which controls the *vertical position* of the pole image, not the *automorphy factor* $|c\tau_p + d|$ driving the divergence. So s is the wrong regulator: it does not damp the (c, d) sum. (One genuine subtlety: the residue phases $(c\tau_p + d)^{k-2}$ oscillate, so conditional summation / analytic continuation is not a priori hopeless—but the continuation that works is Hecke’s trick, not the s -variable.)

(b) Duncan–Frenkel Rademacher-sum regularization — wrong failure mode. Duncan–Frenkel (*Rademacher sums, moonshine and gravity*) regularize sums whose *seeds are cusp polar parts* (q^{-m} at $i\infty$), with the delicate case being weight ≤ 0 / conditional convergence. For positive weight their sums are just the holomorphic Poincaré series, absolutely and locally uniformly convergent for weight > 1 —cusp seed, no interior poles. The divergence here is different in kind: interior-pole seeds whose summands grow in *modulus* like $|c\tau_p + d|^{k-2}$ (absolute divergence, not a conditional/phase problem). Their truncate-and-correct machinery has no purchase on that growth. The correct precedent is again Hecke’s trick à la Bringmann–Kane (weight 0). Duncan–Frenkel is the right tool for the (already solved) negative-weight side, not the open positive-weight side.

5 The inversion, and the LHMf / Löbrich–Schwagenscheidt comparison

The inversion. Rather than form the divergent orbit sum, take the finite-contour period/ L -function as the regularization-free invariant the divergent sum was only ever a proxy for, and let the wall-crossing residues track the chamber-to-chamber jumps. The question is whether this is new.

What the literature already contains. *Locally harmonic Maass forms* (Bringmann–Kane–Kohnen, IMRN 2015): functions harmonic except across geodesics, where they jump, becoming continuous only after adding an explicit *local polynomial* depending on the chamber. *Löbrich–Schwagenscheidt* (arXiv:2006.10549, *Locally harmonic Maass forms and periods of meromorphic modular forms*) construct $\mathcal{H}_{1-k,n}(\tau)$ of weight $2 - 2k$ as the n -th *period* (in z) of Petersson’s two-variable kernel

$$H_{k,k-1}(z, \tau) = \sum_{M \in \Gamma} v^{2k-1} (z - \tau)^{-1} (z - \bar{\tau})^{1-2k} \big|_{2-2k, \tau} M,$$

which in z is a weight- $2k$ meromorphic modular form with poles exactly at the Γ -translates of τ . Their Theorem 1.3 splits $\mathcal{H}_{1-k,n}$ into holomorphic and non-holomorphic Eichler integrals plus a

locally polynomial part $\mathcal{P}_{1-k,n}(\tau)$, harmonic off $E_1 = \Gamma \cdot i\mathbb{R}_+$ (translates of the imaginary axis), with the chamber-dependent jump given by the *finite* sum

$$\frac{i^{1-n}}{2} \sum_{\substack{M=\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma \\ ac > 0, \operatorname{Re}(M\tau) < 0}} (\tau^n - (-1)^n \tau^{2k-2-n})|_{2-2k} M.$$

As an application they obtain finite rational formulas for periods of the meromorphic forms $f_{k,P}$ attached to positive definite binary quadratic forms (poles at CM points, weight $2k > 0$).

Term-by-term comparison with DRB. Write the DRB weight as w (so $w = 2k$ in LS notation; period-monomial degree $w - 2 = 2k - 2$).

- **Shared / occupied — the wall-crossing local polynomial.** DRB’s per-crossing residue jump is $2\pi i \sigma \operatorname{Res}(f, \tau_p) (X - \tau_p Y)^{w-2}$. LS’s jump terms are the period-monomials $\tau^n - (-1)^n \tau^{w-2-n}$ slashed by the finitely many M whose wall has been crossed. These are the *same graded object*: DRB bundles the monomials τ_p^ℓ into the single generating polynomial $(X - \tau_p Y)^{w-2}$; LS exposes them componentwise. The chamber/wall-crossing phenomenon is therefore *not new to DRB*.
- **Weaker-than-hoped: “general f .”** The Petersson kernel $H_{k,k-1}(z, \tau)$ already handles an *arbitrary* single pole orbit (any τ , not just CM), with higher order accessed by τ -derivatives; multiple orbits are linear combinations. So generality of the pole configuration is not by itself a strong distinction.
- **Transverse — not present in LS.**
 - (i) The completed *all- s* L -function $L^*(f, s)$. LS computes the *integer* periods r_n ($0 \leq n \leq 2k - 2$)—period-polynomial coefficients—not an L -function over $s \in \mathbb{C}$.
 - (ii) The *quasi-periods* $\eta^\pm$ and the de Rham–Betti period matrix. LS is Betti-side periods only; there is no Brown-style quasi-period / de Rham–Betti coordinatization.
 - (iii) The *finite contour between interior basepoints*. LS’s period is a Mellin integral $\int_0^\infty \dots y^n dy$ anchored at the cusps (Cauchy principal value on walls); DRB never touches the cusp.

Verdict. The overlap is essentially confined to the wall-crossing phenomenon, which is independently known and which the construction itself treats as structural rather than headline. The finite-contour extraction of the completed L -function and the joint period/quasi-period reconstruction are transverse to the LHMF / cycle-integral line. The observation that the Bringmann–Schwagenscheidt cycle-integral papers do not cite 1806 is consistent with this: their deliverable is cycle-integral *periods* of $f_{k,P}$ via Shintani lifts, a different object from the completed L -function of a general interior-pole meromorphic form. “Not cited” is weak evidence on its own; the substantive argument is the difference in deliverable, and that argument holds.

6 Novelty ledger

Occupied (do not claim). The chamber / wall-crossing local-polynomial structure for periods of meromorphic forms (Bringmann–Kane–Kohnen; Löbrich–Schwagenscheidt; the pole-bearing period *functions* of Choie–Zagier; the cohomological framework of Bruggeman–Choie–Diamantis).

Negative-weight and weight-0 coefficient formulas (Petersson; Bringmann–Kane; HR 1919). Positive-weight coefficients for the special family $f_{k,D}$ via theta lifts (Löbrich–Schwagenscheidt arithmetic; Li–Neururer; Bengoechea).

Plausibly novel (not found pre-empted; must be positioned). The finite-contour de Rham–Betti extraction of $\omega^\pm$ and $\eta^\pm$ for general meromorphic f ; the all- s $L^*(f, s)$ carrying explicit wall-crossing; and the proof that the BFK incomplete-gamma L -function is the correct finite object, with an explicit numerically convergent reconstruction.

Still open (delivered by none of the above). A convergent positive-weight Rademacher *coefficient* formula for $c_f(n)$ of general meromorphic forms regular at the cusp. The obstruction is the $|c\tau_p + d|^{k-2}$ growth; the only known route is Hecke-trick analytic continuation, and it has not been carried out in this generality.

7 Priority and prior art (forward-citation sweep)

A forward-citation sweep of BFK and 1806, filtered for any finite-contour or de Rham–Betti extraction of the completed L -function, returns the following.

- The completed L -function of an interior-pole or weakly holomorphic form is by now a small but real subfield: BFK (weakly holomorphic); 1806 (meromorphic, regular at the cusp); Löbrich–Schwagenscheidt (L -values of special meromorphic forms via Cauchy principal value); Wang–Zhang, *Meromorphic quasi-modular forms and their L -functions* (2022), via regularized integrals; and a completed L -function for a weight-2 polar harmonic Maass form (arXiv:2201.03146). Hence “first L -function of a meromorphic form” is not an available claim, and $L^*(f, s)$ as a value is the same completed object these define.
- The 1806 paper *is* cited within this lineage: Wang–Zhang reference it (as McGady) for “ L -functions for meromorphic modular forms holomorphic at infinity.” By contrast the LHMF / cycle-integral papers (Bringmann–Kane–Kohnen; Löbrich–Schwagenscheidt; Alfes–Neumann–Bringmann–Schwagenscheidt) do not cite it, consistent with their deliverable being cycle-integral periods rather than the completed L -function. Cited where the object coincides, not cited where it differs—the pattern is itself evidence of transversality, not oversight.
- Every paper in the L -function niche defines the L -function through a *regularized* Mellin integral from 0 to $i\infty$ (cusp regularization and/or Cauchy principal value at poles). None performs the finite-contour cocycle extraction between interior basepoints; none extracts quasi-periods $\eta^\pm$ alongside periods $\omega^\pm$; none frames the result in Brown’s de Rham–Betti period structure. The priority window on the *method* and on the *joint period/quasi-period/ L -function package* therefore appears clean.

8 Recommended framing and proposed title

Lead with the L -function. The distinctive content is *uniformity*: a single construction whose completed L -function is the same object across cuspidal, weakly holomorphic, and interior-pole forms—regimes that previously required mutually incompatible tools. BFK’s regulator is regime-locked (it leans on the harmonic-Maass / non-holomorphic completion and has no purchase on interior poles); the finite-contour deformation handles interior poles and also processes $\hat{\Delta}$ with

the identical kernel; Brown’s cocycle yields $\omega^\pm, \eta^\pm$ but no L -function. Nobody has run all of this through one object. The L -function is the right lead object precisely because it is what stays fixed while the input ranges over the three regimes.

The claim to make: a single finite-contour cocycle yielding the completed all- s L -function *together with* the periods $\omega^\pm$ and quasi-periods $\eta^\pm$, uniformly across cuspidal, weakly holomorphic, and interior-pole forms, with nothing to regulate.

The phrasing to avoid: “first L -function for both weakly holomorphic and meromorphic forms.” Wang–Zhang’s regularized integral arguably already spans cusp-exponential growth and interior poles within one formalism, so they can contest that exact priority. What they lack is the rest of the bundle—no periods, no quasi-periods, no cocycle, regularized rather than finite. State the unification as the cocycle bundle, not as priority over Wang–Zhang on the bare L -function.

Proposed manuscript title (replacing the current working title):

Finite-contour cocycles for (quasi-)periods and L -functions of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$.

“Finite-contour cocycles” puts the method—the part with no other occupant—up front; “(quasi-)periods” surfaces the $\eta^\pm$ / de Rham–Betti differentiator while folding in $\omega^\pm$; “ L -functions” stays in the lead per the uniformity argument above; “on $\mathrm{SL}_2(\mathbb{Z})$ ” is honest level-1 scoping that pre-empts the higher-level reflex. Read maximally, “meromorphic” subsumes the weakly holomorphic and cuspidal cases, so the span lives in the title; the abstract’s opening must then make “one construction, all three regimes, L and (quasi-)periods together” load-bearing from the first line, or a skimmer files the paper under the occupied L -function niche.

Leaner fallbacks: *Finite-contour cocycles for (quasi-)periods and L -functions of meromorphic modular forms*; or *(Quasi-)periods and L -functions of meromorphic modular forms via finite-contour cocycles*.

9 Recommended actions

1. Retitle the main manuscript to the proposed title in §8, and make the uniformity claim (one cocycle; the L -function with $\omega^\pm, \eta^\pm$; three regimes) load-bearing in the first two sentences of the abstract.
2. Cite Bringmann–Kane–Kohnen (2015) and Löbrich–Schwagenscheidt (2020) in the DRB note; they are the closest references and are currently absent.
3. Position the residue jump $(X - \tau_p Y)^{w-2}$ explicitly against the LS local polynomial, and test whether, for poles at CM points, the DRB pole-crossing residues are the specialization of Choie–Zagier rational period functions. This either claims or disclaims the point-crossing structure cleanly.
4. Frame the contribution as “finite-contour de Rham–Betti L -function and quasi-period reconstruction” (transverse to LHMF/LS), *not* as a new wall-crossing phenomenon and *not* as a positive-weight coefficient formula.
5. If pursuing positive-weight coefficients: the regulator is Hecke’s trick (as in the Bringmann–Kane weight-0 Petersson problem), not the DRB s -variable and not Duncan–Frenkel.

Key references

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